

Research Assignment Answers

Data Foundations

1. A database is a place where information is stored and organised so that it is easy to find, change or use it.

For example, a bank uses database to keep a customers history of transactions as well as account information. Another example is online shops like Take a lot, they use database to store your information and records of purchase and later use the same data to send you personalized emails based on your historical purchases.

Source : IBM-Whats Database

Ans.

2. A data warehouse is a central place where data from different sources is brought together for reporting & analysis, before it is loaded on the data warehouse it goes through ETL process so that it is ready for analysis.

A regular database is mainly for storing or used for everyday activities such as transactions at a store while a warehouse is more focused on analysis of information. A company would use a data warehouse when it wants to combine data from different systems and use it for analysis & make proper business decisions.

source : IBM - What is datawarehouse

3. A data lake is a storage system that can hold large amounts of data in its original form. That data may be structured (in rows & columns) or unstructured (images, videos, pdfs --)

The difference between a data warehouse and data lake is that, data lake can store data in its raw state while a ^{data} warehouse stores organised data that is ready for analysis.

If a business has large data containing different datatypes and doesn't know how the data will be used as yet, they may prefer to use a data lake.

Source : Microsoft Azure

4.

Data mart is a smaller part of a data warehouse that focuses on one particular area of a business. For example a business could have a sales data mart for sales department or a finance data mart for finance department in the same company making it easy for employees to work with information that is relevant to them.

Source : IBM

5. • Structured data $\Rightarrow$ organised in a fixed format. (rows & columns)

e.g. customer table in a database

• Semi-structured data $\Rightarrow$ doesn't use normal table structure but organised. e.g. a JSON file

• Unstructured data $\Rightarrow$ doesn't have fixed structure. e.g. images, videos, audios

Source : IBM

6. Data modelling is a process of deciding how data should be organized & how different parts / pieces of data should relate to each other.

It is important when working with data as it makes it easier to store, find & analyse data.

- Relational model → organises data into related tables
- Dimensional model → used for reporting & data warehouse.

source: IBM

7. Data mining is the process of looking through large amount of data to find useful patterns or information. Businesses can use it to understand customers by looking at their purchasing patterns, detect fraud and find some trends. For example a store can analyse customer purchases (eg Take a lot ^{or ubereats} has that suggestion that you must take another particular item because they are often bought together). Banks can also detect unusual activity or transactions that could indicate fraud and send you messages or block account.

source: IBM

8. DBMS is a software that allows people to store, organise, update & retrieve data from databases

- MySQL → commonly used for websites & applications
- PostgreSQL → used for complex relational database applications
- Oracle → used often by large organisations with large amounts of important data
- MongoDB → requires no SQL (NoSQL) database used when data has a more flexible structure.

Sources: IBM, About PostgreSQL (PostgreSQL), MongoDB - About MongoDB

9. ETL (Extract Transform Load)

E → extract / collect data from different sources

T → cleaning & changing the data into useful format

L → putting the prepared data into a destination such as a data warehouse.

ETL is important in data analytics as analysts need to clean and organise data before they can produce reliable results.

Sources : Brightlearn Introduction to Database pdf, IBM

10. CSV → stores data in rows & columns and useful for simple datasets

JSON → stores data using keys and values and is commonly used with APIs, easy for humans to read.

XML → similar to HTML, it uses tags to describe data

~~Parquet~~ Parquet → column based format that ^{is} useful for large datasets & analytics as it can store data efficiently.

Avro → used for data exchange & streaming systems

Sources : devoto, Parquet.apache.org

11. Artificial Intelligence (AI) is a technology that allows computers to perform tasks that normally require human intelligence.

- Narrow AI (weak) → designed to perform specific tasks e.g. voice assistant that can answer questions & perform commands

- General AI (strong) → performs different intellectual tasks like a human

Sources : IBM-AI

12. Machine Learning is a part of AI where computers learn patterns from data instead of being given instructions for every possible situation. In traditional programming human write explicit rules and input data into a computer to produce an answer

- Supervised learning → the model learns from data that already has correct answers
- Unsupervised learning → model looks for patterns in data without given answers
- Reinforcement learning → system learns by receiving rewards or penalties for its actions

Sources: scribbr.com, Google - Machine Learning Crash Course

13. Deep learning is a type of Machine Learning that uses neural networks with several layers to learn complicated patterns

↳ interconnected nodes arranged in layers.

Information moves through the network &

(AI → ML → DL)

the system learns which patterns are important for producing output

Sources: IBM

↳ Natural Language Processing

14. NLP allows computers to work with human language (e.g. voice assistants, chatbots & search engines) they use NLP to understand what we say or type.

Source: IBM

15. LLM is an AI model that is trained on a very large amount of text so that it can understand & generate human-like languages. Machine learning model may be built for a particular task while LLM can perform different language tasks.

Source: Google Gemini, Open AI

16. Generative AI is AI that can create new content based on patterns it has learned from existing data.
e.g. ChatGPT → can generate & work w/ text
• Google Gemini → can generate content & answer questions
• Image-generative AI → can create an image from simply reading your written description of what you want

Source: IBM

17. Training dataset is all examples that are used to test or train an AI or machine learning model. If the data given is incorrect/biased the model can learn these problems and produce unreliable results as it learns from information given.

Source: Google - Data characters

18. Computer vision is an area of AI that allows computers to analyse and understand images & videos. It doesn't see like humans when seeing an image. It processes the information in the image and uses patterns to identify objects or features.

For example facial recognition and self-driving cars

Source: TBM - what is computer vision

19. API allows two different software systems to communicate with one another. It is like "middle man".

They are important as they allow different applications to share information and work together.

Source: Postman - what is an API

20. API key is a code used to identify or ~~or~~ authenticate an application when it accesses an API and it should be protected in the same way as a password as someone else ~~who~~ would be able to use the API under your account if it's shared, that could result in unwanted changes or access to confidential information.

Source: Postman-What is an API key

21. REST stands for Representational State Transfer. REST API allows systems to communicate over the web using HTTP.

Main HTTP methods:

- GET - gets information
- POST - creates new information
- PUT - updates existing information
- DELETE - removes information

Source: MDN-HTTP Methods

22. JSON is a format used to store and exchange data. It is commonly used when working with APIs as it is easy for people to read and easy for computers to process

eg

```
{ "name": "Lisoletu",  
  "occupation": "Data Analyst"  
}
```

Source: MDN, JSON

23. Prompt engineering means writing clear instructions for an AI tool so that it understands what I want it to do. It's important as a vague prompt would result in a vague answer.

Examples of poor prompt:

① - "Give me a recipe" (poor prompt)

- Give me a 15-minute, high protein chicken recipe that uses fewer than 5 ingredients (better prompt)

② - "Make my portfolio" (poor prompt)

- "use my cv to create a professional portfolio about me, do not invent skills or anything use the information from my cv only" (better prompt)

Source: Open AI. Prompt Engineering Best Practices

24. A MAP function applies the same operation to each item in a list and returns the results.

e.g. in Python:

```
numbers = [1, 2, 3, 4]
```

```
result = list(map(lambda x: x * 2, numbers))
```

↓

[2, 4, 6, 8] NB The function takes each number & multiply it by 2.

Source: Python Software Foundation (Built-in Functions - map()).

25. Cloud computing means using computing services such as storage, servers and software through the internet instead of having to manage all the physical equipment yourself.

The 3 main models:

- IaaS → provides infrastructure such as virtual machines, e.g. Azure Virtual machines

- PaaS - provides a platform for developing and running applications

e.g. Azure App Service

- SaaS → provides ready-to-use software over the internet
e.g. Microsoft 365.

Source: Microsoft Azure, What is cloud Computing?

26. A Version Control System is a system that keeps track of changes made to files over time. & This means I can see previous versions of my work and return to an earlier version if something goes wrong.

Git is widely used version control system and helps data analysts and developers to manage ^{their} ~~the~~ code and collaborate on projects.

- Commit ⇒ is a saved set of changes
- branch ⇒ is a separate version of a project where I can work on changes without immediately affecting the main version

Source: Git, About Version Control

27. A Jupyter Notebook is an interactive document where I can write and run code, see the results and add explanations in the same place. It is commonly used with Python and is popular with data analysts & data scientists because it can be used for data cleaning, analysis, calculations and visualizations -

Source: Project Jupyter, Jupyter

28. Python is a programming language & is considered most popular because its syntax is easy to understand and it has many libraries for data & AI.

4 common libraries :

① Pandas → used for working with & analysing tables of data

② Numpy → used for numerical calculations

③ Matplotlib → used to create graphs & charts

④ Scikit-learn → used for machine learning

Sources : Python.org - About Python, Pandas Getting started

Numpy Documentation, Matplotlib Documentation,

Scikit-learn, Machine learning in Python

29. Data privacy is about protecting people's personal information and making sure it is collected & used responsibly.

GDPR (General Data Protection Regulation) is a European Union data protection law that gives people rights over their personal information and places requirements on organisations that process it.

SOURCE : European Commission - GDPR

30. Data ethics is about making sure that data and AI are used responsibly and fairly.

Examples :

① Using someone's personal information without permission can invade their privacy

② Using biased data to make decisions which can affect ~~some~~ certain people.

③ Using AI to spread false information which can be misleading to people

SOURCE : OECD - AI principles

31. Data governance is a set of rules, responsibilities and processes used by an organisation to manage its data properly. Some important parts include data quality, security, privacy, ownership, access controls & policies

Source: IBM - Data Governance

32. AI bias happens when an AI system produces unfair or less accurate results for certain people/groups. That can be caused by bias training data. For example, facial recognition systems can have different levels of accuracy across different races of people.

Source: NIST - AI Fundamental Research (Managing AI Bias)

33. Role	What they do (Jobs)	Skills & tools
Data Analyst	Looks at data and create insights & reports	SQL, Excel, Power BI, Tableau
Data Scientist	Uses statistics and machine learning to find patterns and make predictions.	Python, statistics ^{statistics} , machine learning
Data Engineer	Builds systems and pipelines that collect and prepare data	SQL, Python, Databricks, Cloud tools
AI Engineer	Builds & put AI systems into applications	Python, machine learning, APIs & AI tools

Sources: IBM - Data Careers

34. Business Intelligence (BI) means using data and technology to understand how a business is performing and support better decisions.

Two popular BI tools: ~~Power~~ Power BI & Tableau (They allow users to connect to data, create visuals & build dashboards).

For example a business could use a dashboard to monitor sales, revenue and customer trends instead of looking through large tables with numbers.

Source: Microsoft Power BI - Business Intelligence

35. The most interesting concept for me was AI & Machine Learning. I found it interesting because I realised that AI is used for much more than just robots, it can analyse data & recognise patterns, understand language & make predictions.

36. Understanding the difference between data lake, warehouse & mart was quite challenging at first for me as they seemed very similar to one another. After some research I now understand that data lake can also store raw data while warehouse stores organised data and data mart is a smaller section focused on a specific department in a business.

Resources that helped me:

- IBM - What is data warehouse?
- IBM - What is data mart?
- Microsoft Azure - Data Lake.

37. Data & AI will definitely have a big impact on my future career. I can use data to find trends, create dashboards and help businesses make better decisions. AI can help me with tasks such as coding, research and analysing information. AI is already part of everyday life as you can literally search anything e.g. how specific tool works etc.

References

1. IBM Databases IBM Think
www.ibm.com/think/topics/database
2. IBM What is Datawarehouse IBM Think
www.ibm.com/think/topics/datawarehouse
3. Microsoft, What is Data Lake? Microsoft Learn
learn.microsoft.com/en-us/azure/architecture/data-guide/scenarios/datalake
4. IBM (2021). What is ETL? IBM Think
www.ibm.com/think/topics/etl
5. IBM (2024). What is Artificial Intelligence. IBM Think
ibm.com/think/topics/artificial-intelligence
6. Google. Machine Learning Crash course, Google For Developers
developers.google.com/machine-learning/crash-course
7. IBM (2025). What is Deep Learning IBM Think
www.ibm.com/think/topics/deep-learning
8. IBM (2024). What is NLP, IBM Think
www.ibm.com/think/topics/natural-language-processing
9. Postman. What is API? Postman
www.postman.com/what-is-an-api/
10. Git, About Version Control, Git
git-scm.com/book/en/v2/Getting-started-About-Version-Control

11. Project Jupyter

jupyter.org

12. Python Software Foundation, Python documentation

www.python.org/doc/

13. ~~Scribbler~~ Dev

dev.to

14. European Commission, General Data Protection Regulation

commission.europa.eu/law/law-topic/data-protection/rules-business-and-organisations/application-regulation

15. OECD (2024) - AI principles

www.oecd.org/en/topics/ai-principles.html

16. NIST - Managing AI bias

www.nist.gov/artificial-intelligence/ai-fundamental-research-managing-ai-bias

17. Microsoft, What is Business Intelligence? Microsoft Power BI

www.microsoft.com/en-us/power-platform/products/power-bi/topics/business-intelligence/what-is-business-intelligence